

answerdotai/**answerai-colbert-small-v1**

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sentence-transformers

ONNX

Safetensors

English

bert

ColBERT

multi-vector

RAGatouille

passage-retrieval

arxiv:2407.20750

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Model card



Files

xet



Community 17

answerai-colbert-small-v1

answerai-colbert-small-v1 is a new, proof-of-concept model by [Answer.AI](#), showing the strong performance multi-vector models with the new [JaColBERTv2.5 training recipe](#) and some extra tweaks can reach, even with just **33 million parameters**.

While being MiniLM-sized, it outperforms all previous similarly-sized models on common benchmarks, and even outperforms much larger popular models such as e5-large-v2 or bge-base-en-v1.5.

For more information about this model or how it was trained, head over to the [announcement blogpost](#).

Usage

This model was designed with the upcoming RAGatouille overhaul in mind. However, it's

Downloads last month

384,196

Safetensors

Model size 33.4M params

Tensor type F32

Files info

Inference Providers **NEW**

This model isn't deployed by any Inference Provider.



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Model tree for answerdotai/answe...

Finetunes 5 models

 Quantizations [2 models](#)

Spaces using answerdotai/ans... 14

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compatible with all recent ColBERT implementations!

Sentence Transformers

This model can be used with [Sentence Transformers](#) as a multi-vector (ColBERT-style late interaction) retriever via the `MultiVectorEncoder`:

```
pip install "sentence-transformers>=6.0.0"

from sentence_transformers import MultiVectorEncoder

model = MultiVectorEncoder("answerdotai/answerai-ja-colbertv2.5")

query = "Which planet is known as the Red Planet?"
documents = [
    "Venus is often called Earth's twin because of its similar size and proximity to Earth.",
    "Mars, known for its reddish appearance, is the second planet from the Sun and the fourth planet from Earth.",
    "Jupiter, the largest planet in our solar system, is a gas giant with a prominent equatorial band.",
    "Saturn, famous for its rings, is sometimes called the 'Jewel of the Solar System'."
]

query_embeddings = model.encode_query(query)
document_embeddings = model.encode_documents(documents)

print(query_embeddings.shape, document_embeddings.shape)
# (32, 96) (17, 96)

# MaxSim late-interaction scoring (higher is better)
scores = model.similarity(query_embeddings, document_embeddings)
print(scores)
# tensor([[30.5692, 31.4895, 31.3029, 31.3071]])
```

Rerankers

If you're interested in using this model as a reranker (it vastly outperforms cross-encoders its size!), you can do so via the [rerankers](#) library:

 [arjun10g/rag-architecture-advisor](#)

 [ashishanand/car_manual_assistant](#)

 [davidr70/ragtou_test](#)

 [JeanCGuerrero/Question_Answerin...](#)

+ 6 Spaces

 **Collection including** [answerdotai/...](#)

Multi-Vector Retrievers  Collection
2 items • Updated Aug 20, 2024 •  4

 **Paper for** [answerdotai/answerai-...](#)

JaColBERTv2.5: Optimising Multi-V...
 Paper • 2407.20750 • Publi... •  22

```
pip install --upgrade rerankers[transformers]
```

```
from rerankers import Reranker
```

```
ranker = Reranker("answerdotai/answerai-colbert-v2")  
docs = ['Hayao Miyazaki is a Japanese director and animator']  
query = 'Who directed spirited away?'  
ranker.rank(query=query, docs=docs)
```

RAGatouille

```
pip install --upgrade ragatouille
```

```
from ragatouille import RAGPretrainedModel
```

```
RAG = RAGPretrainedModel.from_pretrained("answerdotai/ragatouille")
```

```
docs = ['Hayao Miyazaki is a Japanese director and animator']
```

```
RAG.index(docs, index_name="ghibli")
```

```
query = 'Who directed spirited away?'  
results = RAG.search(query)
```

Stanford ColBERT

```
pip install --upgrade colbert-ai
```

Indexing

```
from colbert import Indexer
```

```
from colbert.infra import Run, RunConfig, ColbertConfig
```

```
INDEX_NAME = "DEFINE_HERE"
```

```
if __name__ == "__main__":
    config = ColBERTConfig(
        doc_maxlen=512,
        nbits=2
    )
    indexer = Indexer(
        checkpoint="answerdotai/answerai-collbert-v1.0",
        config=config,
    )
    docs = ['Hayao Miyazaki is a Japanese director, animator, producer, screenwriter, and writer of animated films, manga, and books. He is best known for his Studio Ghibli films, which have won numerous awards and are beloved by audiences worldwide. Miyazaki is also known for his work on the anime series "Castle in the Sky" and "The Wind Named Amiko".']
    indexer.index(name=INDEX_NAME, collection=docs)
```

Querying

```
from colbert import Searcher

from colbert.infra import Run, RunConfig, ColbertConfig

INDEX_NAME = "THE_INDEX_YOU_CREATED"

k = 10

if __name__ == "__main__":
    config = ColbertConfig(
        query_maxlen=32 # Adjust as needed,
    )
    searcher = Searcher(
        index=index_name,
        config=config
    )
    query = 'Who directed spirited away?'
    results = searcher.search(query, k=k)
```

Extracting Vectors

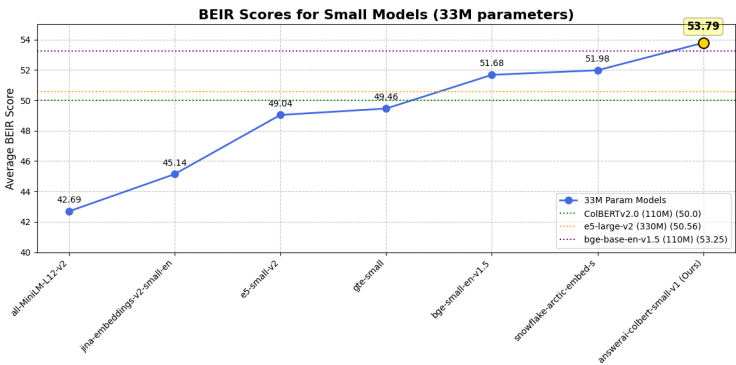
Finally, if you want to extract individual vectors, you can use the model this way:

```
from colbert.modeling.checkpoint import Checkpoint
```

```
ckpt = Checkpoint("answerdotai/answerai-colbert-  
embedded_query = ckpt.queryFromText(["Who dul
```

Results

Against single-vector models



Dataset / Model	answer- colbert- s	snowflake- s	bge- small en
Size	33M (1x)	33M (1x)	33M (1x)
BEIR AVG	53.79	51.99	51.68
FiQA2018	41.15	40.65	40.34
Hot pot QA	76.11	66.54	69.94
MSMARCO	43.5	40.23	40.83
NQ	59.1	50.9	50.18
TRECCOVID	84.59	80.12	75.9
ArguAna	50.09	57.59	59.55
ClimateFEVER	33.07	35.2	31.84
CQADupstackRetrieval	38.75	39.65	39.05
DBpedia	45.58	41.02	40.03

Dataset / Model	answer-colbert-s	snowflake-s	bge-small-en
FEVER	90.96	87.13	86.64
NFCorpus	37.3	34.92	34.3
QuoraRetrieval	87.72	88.41	88.78
SCIDOCS	18.42	21.82	20.52
SciFact	74.77	72.22	71.28
Touche2020	25.69	23.48	26.04

Against ColBERTv2.0

Dataset / Model	answerai-colbert-small-v1	ColBERTv2.0
BEIR AVG	53.79	50.02
DBPedia	45.58	44.6
FiQA2018	41.15	35.6
NQ	59.1	56.2
Hotpot QA	76.11	66.7
NFCorpus	37.3	33.8
TRECCOVID	84.59	73.3
Touche2020	25.69	26.3
ArguAna	50.09	46.3
ClimateFEVER	33.07	17.6
FEVER	90.96	78.5

Dataset / Model	answerai-colbert-small-v1	ColBERTv2.0
QuoraRetrieval	87.72	85.2
SCIDOCS	18.42	15.4
SciFact	74.77	69.3

Referencing

We'll most likely eventually release a technical report. In the meantime, if you use this model or other models following the JaColBERTv2.5 recipe and would like to give us credit, please cite the JaColBERTv2.5 journal pre-print:

```
@article{clavie2024jacolbertv2,  
  title={JaColBERTv2.5: Optimizing Multi-Grained  
  author={Clavi{\`e}, Benjamin},  
  journal={arXiv preprint arXiv:2407.20750},  
  year={2024}  
}
```